PROGRAMMING 1 WARM-UP EXERCISES

Tasks to do in class

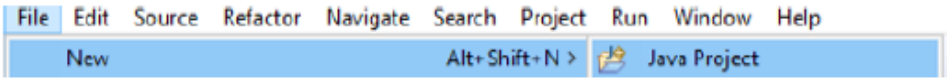
1. In Eclipse, create a new Java Project named **Programming_1**

2. In the **Programming_1** project, create the following three packages:
   - **warmup**
   - **week_1**
   - **week_2**

3. Complete the Java programming exercises that are available on the next page.

ECLIPSE INSTRUCTIONS

How to create a new Java Project



File > New > **Java Project** > *Enter the project name* > **Next** > *Unselect 'Create module-info.java file '* > **Finish**

NB! If you do not see **Java Project** as the first item in the drop down list as above, then you must change the current perspective to **Java Perspective** as follows: **Window** > Open Perspective > **Java**

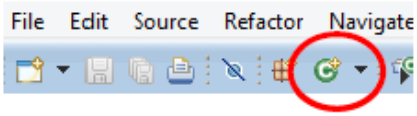
* How to create a new package in the project (*optional*)

Right-click on the **src** folder name or project name > **New** > **Package**

* How to create a new Java program

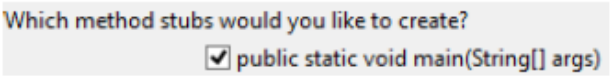
1. Right-click on the package name (or the **src** folder name, or the project name) > **New** > **Class**

   OR Click on the following icon:



Then,

2. Enter the name of the class to be created
3. **Select** the method **public static void main(String[] args)** to be created. Unselect the other options.



4. Finally, click on **Finish**

* How to run a Java program

Alternative ways of running the Java program in the active tab of the editor include the following:

* Run > **Run**
* Press **Ctrl + F11**
* Click on the Run icon

NOTE: If there is a syntax error in your code, Eclipse displays the error message in the **Problems** window.

JAVA SYNTAX DETAILS

In this warm-up exercise set, **you can apply the JavaScript syntax presented in the previous course** with the following differences:

1. You should write your own Java code ***within the main method*** in the program class.

2. Use *System.out.println* instead of *console.log* to print to the Console window.

```
System.out.println("Hello!");
```

3. To define a variable, use the keyword *int* instead of *var* or *let*.

```
int x = 0;
```

4. Create an array of integers as follows:

```
int[] someNumbers = {1, 2, 6, 9};
```

IMPORTANT!

→ To insert `System.out.println()` quickly, type **syso**, then press **Ctrl+Space**

→ To format your code correctly, press **Ctrl+Shift+F**

WARM-UP EXERCISES (Not to be submitted !)

Create these programs in the ***warmup*** package in the ***Programming_1*** project.

1. Create a Java program **WarmUp_1** that prints your name.

2. Create a Java program **WarmUp_2** that prints the numbers from 1 to 10, each on a new line.
   - Use a ***for loop*** to iterate through the integers and the **System.out.println** method to display each number.

3. Create a Java program **WarmUp_3** that prints the odd numbers from 1 to 11, each on a new line.
   - This time, use a ***while loop*** to iterate through the integers and the **System.out.println** method to display each number.

4. Create a Java program **WarmUp_4** that prints even numbers from 2 to 20. If the number is 10, the program should print "Ten" instead of the number.
   - Use a single ***for loop*** to iterate through the numbers and an ***if-else*** statement to handle the special case for the number 10.

5. Create a Java program **WarmUp_5** that calculates and prints the sum of all integers in an array.
   - The array should contain the following integers: 1, 2, 3, 16, 20
   - Use a ***for loop*** to iterate through the array